

assignment torques

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:01pm

Quiz Instructions

There is no tutorial. I made an easier assignment for you



Question 1 10 pts

If Alex wishes to rotate his skateboard, then he must apply a

☐

pause before bearing down on the board.

☐

torque.

☐

rotational maneuver.



Question 2 10 pts

A 15-kg child is sitting on a playground teeter-totter (see_saw), 1.5 m from the pivot. What is the magnitude of the minimum force, applied 0.30 m on the other side of the pivot, that is needed to make the child lift off the ground? (about)

☐

44 N

☐

23 N

☐

66 N

☐

75 N

☐

740 N



Question 3 10 pts

The figure shows a person's foot. In that figure, the Achilles tendon exerts a force of magnitude $F = 720$ N. What is the magnitude of the torque that this force produces about the ankle joint?

(see my side or your textbook or you can go to 10:45 min in this video:

[https://www.youtube.com/watch?](https://www.youtube.com/watch?v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2)

[v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2](https://www.youtube.com/watch?v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2) 

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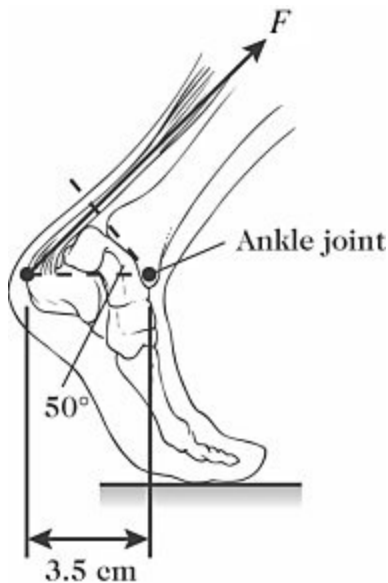
[v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2\)](https://www.youtube.com/watch?v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2)



[\(https://www.youtube.com/watch?](https://www.youtube.com/watch?v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2)

[v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2\)](https://www.youtube.com/watch?v=LssKoGfjB1Q&list=PLkRMGCE8BEtZrQkaAfojaqjQVo8CQPLW2)

)



☐ 16 N · m

☐ 26 N · m

☐ 12 N · m

☐ 21 N · m

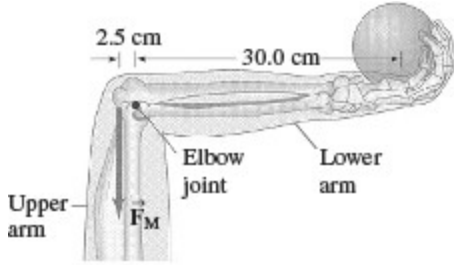
☐ 36 N · m



Question 4 10 pts

An athlete holds a 7.5-kg shot put in his hand with his lower arm horizontal, as shown in the figure. His lower arm has a mass of 2.8 kg and its center of gravity (or center of mass) is 12 cm

from the elbow-joint pivot. How much force must the extensor muscle (which is $\vec{F}_M$ in the figure) in the upper arm exert on the lower arm?



- ☐ 1500 N
- ☐ 1000 N
- ☐ 500 N
- ☐ 100 N



Question 5 10 pts

If you place a pipe over the end of a wrench when trying to rotate a stubborn bolt, effectively making the wrench handle twice as long, you'll multiply the torque by

- ☐ four.
- ☐ two.
- ☐ eight.



Question 6 10 pts

A man in a gym is holding an 8.0-kg weight at arm's length, a distance of 0.55 m from his shoulder joint. What is the torque about his shoulder joint due to the weight if his arm is held at 30° below the horizontal?

- ☐ 22 N · m
- ☐ 2.2 N · m
- ☐ 37 N · m
- ☐ 4.4 N · m



13 N · m



Question 7 10 pts

A man in a gym is holding an 8.0-kg weight at arm's length, a distance of 0.55 m from his shoulder joint. What is the torque about his shoulder joint due to the weight if his arm is horizontal?



43 N · m



0 N · m



0.24 N · m



15 N · m

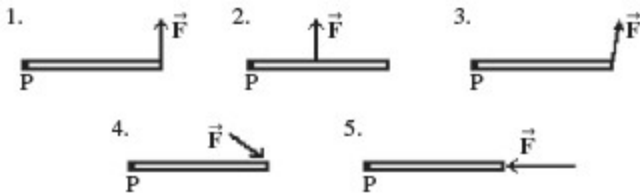


4.4 N · m



Question 8 10 pts

As shown in the figure, a given force is applied to a rod in several different ways. In which case is the torque about the pivot P due to this force the greatest?



1



2



3



4



5



Question 9 10 pts

Two people are balanced on a seesaw. If one person leans inward toward the center of the seesaw, that person's end of the seesaw tends to

☐

rise.

☐

stay at the same level.

☐

fall.

☐

need more information



Question 10 10 pts

On a balanced seesaw, a boy three times as heavy as his partner sits

☐

more than $\frac{1}{3}$ the distance from the fulcrum.

☐

less than $\frac{1}{3}$ the distance from the fulcrum.

☐

$\frac{1}{3}$ the distance from the fulcrum.

Not saved

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